

Boundary Conditions and Open or Closed Endpoints

Answer Key

Algebra 1

Quick Answers

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|--|---|
| 1. 9 | 6. (b) the cost of a 2-hour stay = 6, (c) the cost of a 3-hour stay = 12, — |
| 2. $f(2) = 5$, $f(5) = 7$ | 7. 3 |
| 3. (b) = 14, — | 8. (a) the value of $a = 2$, (b) $q(-1) = 1$, (b) $q(2) = 4$, — |
| 4. (a) solutions of the formula = -5 , 5 , — | |
| 5. -1 | |
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What is verified

4 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 3, 4, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSA-REI.B.4 – *problem 4*

HSF-IF.A – *problems 1, 2, 3, 6*

HSF-IF.B – *problems 5, 7, 8*

Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

1. *If they got 7: used the first rule at the boundary, although its condition is $x < 3$*
 3. *If they got 13: used the rule written for $x > 4$ at the boundary itself*
 5. *If they got 1: matched the two formulas at $x = 0$ instead of at the boundary $x = 2$*
 6. *If they got 8: used the second rule at $t = 2$, which the printed condition excludes*
 7. *If they got 2: dropped the plus 5 from the second formula and solved $5c = 10$*
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1. $f(x) = 2x + 1$ for $x < 3$, and $12 - x$ for $x \geq 3$. Find $f(3)$.

Solution: Read the conditions before the formulas. The first says $x < 3$, which stops just short of 3; the second says $x \geq 3$, which includes it. So 3 belongs to the second rule: $12 - 3$.

9

2. Using the same f , find $f(2)$ and $f(5)$.

Solution: $2 < 3$ takes the first rule; $5 \geq 3$ takes the second.

$$f(2) = 2(2) + 1 = 4 + 1$$

$$f(5) = 12 - 5$$

$$f(2) = 5, \quad f(5) = 7$$

3. Leo computes $g(4) = 3(4) + 1 = 13$, where $g(x) = x^2 - 2$ for $x \leq 4$ and $3x + 1$ for $x > 4$.
(a) Name his error. (b) Find $g(4)$.

(a) — instructor-judged. Leo's arithmetic is correct: $3(4) + 1$ really is 13. The error is the choice of rule. The value 4 satisfies $x \leq 4$, so the *first* rule owns the boundary; the second is written for $x > 4$ strictly, so it says nothing at all about $x = 4$. Full credit requires naming the condition that decides it, not merely recomputing.

Solution (b): $g(4) = 4^2 - 2 = 16 - 2$.

14

4. With g as above and $g(x) = 23$: (a) solve $x^2 - 2 = 23$; (b) which value is genuine?

Solution (a): $x^2 = 25$, and two numbers square to 25.

$$x = -5 \quad \text{and} \quad x = 5$$

(b) — instructor-judged. Only $x = -5$ is a solution of $g(x) = 23$. Both values satisfy the *formula*, but the formula $x^2 - 2$ is only in force when $x \leq 4$. Since $-5 \leq 4$, that value survives; since $5 > 4$, the rule does not apply there and the value must be discarded. Full credit requires rejecting 5 on the condition. Calling 5 an arithmetic mistake is wrong — it does satisfy the equation. (For completeness: on the other branch $3x + 1 = 23$ gives $x = 7\frac{1}{3}$, which does satisfy $x > 4$, so a full investigation finds that solution too; the question here asks only about the first rule.)

5. $h(x) = 3x + k$ for $x < 2$, and $x^2 + 1$ for $x \geq 2$. Find k so there is no jump at $x = 2$.

Solution: “No jump” means the two formulas produce the same value at the boundary, so set them equal *at* $x = 2$ — not at any other input.

$$3(2) + k = 2^2 + 1$$

$$6 + k = 5$$

$$k = -1$$

6. Sofia writes the second condition of $C(t) = 2t + 2$ ($0 \leq t \leq 2$), $4t$ ($t > 2$) as $t \geq 2$. (a) Name the error. (b) $C(2)$. (c) $C(3)$.

(a) — instructor-judged. With $t \geq 2$ on the second rule, both conditions would claim $t = 2$, and the two formulas disagree there: $2(2) + 2 = 6$ while $4(2) = 8$. A price list that returns two different costs for the same two-hour stay is not a function — each input must have exactly one output. Full credit requires naming the overlap *and* the two conflicting values. Saying only “the sign should be $>$ ” with no reason is partial credit.

Solution (b): As printed, $t = 2$ satisfies $0 \leq t \leq 2$, so the first rule applies: $2(2) + 2$.

6

Solution (c): $3 > 2$, so the second rule: $4(3)$.

12

7. $p(x) = cx$ for $x < 5$, and $2x + 5$ for $x \geq 5$. Find c so there is no jump at $x = 5$.

Solution: Equate the two formulas at the boundary $x = 5$, carrying the whole second formula — the constant 5 is part of it.

$$\begin{aligned}c(5) &= 2(5) + 5 \\5c &= 15\end{aligned}$$

$$\boxed{c = 3}$$

8. Synthesis challenge. $q(x) = x + a$ for $x \leq -1$, and x^2 for $x > -1$. (a) Find a . (b) $q(-1)$ and $q(2)$. (c) Which rule gives $q(-1)$, and why?

Solution (a): Set the formulas equal at $x = -1$: $-1 + a = (-1)^2 = 1$.

$$\boxed{a = 2}$$

Solution (b): With $a = 2$: the input -1 satisfies $x \leq -1$, so use $x + a$; the input 2 satisfies $x > -1$, so use x^2 .

$$\begin{aligned}q(-1) &= -1 + 2 \\q(2) &= 2^2\end{aligned}$$

$$\boxed{q(-1) = 1, \quad q(2) = 4}$$

(c) — **instructor-judged.** $q(-1)$ comes from the first rule, $x + a$, because that rule's condition is written with $\leq$ and therefore includes the boundary; the second rule is written for $x > -1$ strictly and does not reach it. Full credit also requires the second half: if both conditions used $\leq$, the input -1 would be claimed by both rules and the definition would hand back two outputs for one input, so it would not define a function. (Choosing $a = 2$ makes the two agree there, which is why the graph has no jump — but agreement is a bonus, not what makes the definition legal.) Answering only that “the value would be different” is not full credit.